
Organisational knowledge creating processes and the performance of university-industry collaborative R&D projects

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Abstract: This paper describes one of the first empirical explorations into organisational knowledge creation theory as first elucidated by Nonaka and Takeuchi [1]. We extend the theory and examine it in the inter-organisational context of University-Industry (U-I) collaborative R&D Projects. More specifically, the relationship between enabling conditions and knowledge conversion processes as well as the effects of these processes on the achievement of technological objectives are studied using a sample of 25 U-I collaborative R&D projects into advanced technology. Quantitative and qualitative evidence supports the theory that some enabling conditions are significant for knowledge conversion processes. Furthermore, the presence of the aggregate knowledge processes is positively associated with the achievement of successful technological objectives. The theoretical and practical implications of these findings are discussed.

Keywords: knowledge creation; collaborative R&D; university-industry collaboration; Nonaka and Takeuchi.

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1 Introduction

The past decade of management research in general has created much interest in knowledge-based theories of the firm [2,3] which account, in part, for the creation of technical knowledge used in achieving a competitive advantage for firms. Similarly, much has been written conceptually regarding general organisational learning processes within firms (e.g., [4–9]). These models of organisational learning reveal how organisations learn at a general level by interpreting their environments and responding appropriately. Whilst these theoretical discourses have been important and influential with regard to organisational adaptation, none of these models speaks directly to technological knowledge creation or deals effectively with the processes by which new organisational knowledge is actually created (e.g., as per R&D activities of the firm). In fact, little theoretical and very little empirical work looks specifically at the creation of new technical knowledge and the underlying processes of turning the unknown into the known at an organisational level, save the well received theory of Nonaka and Takeuchi [1].

Furthermore, the same factors and processes of organisational knowledge creation that take place across organisational boundaries, within the context of interorganisational relationships, are even less well understood. Given that complex and convergent technical innovations often require collaboration to be successful, the challenge is to examine the processes applied to multiple organisations [10]. Recent research suggests that we know little of the processes of knowledge creation, especially in cases involving collaboration between multiple organisations (e.g., [11,12]). Furthermore, when collaborating organisations differ significantly in their environments there is little illuminating research.

In this article we empirically examine the processes employed when multiple organisations collaborate to create technical knowledge. Our theoretical model is derived from the seminal work of Nonaka and Takeuchi [1]. Our focus is on the subset of interorganisational collaborations involving university-industry (U-I) partnerships in advanced technologies. The ultimate objective is to understand better the creative process in the partnered organisations.

The paper begins by examining literature on U-I collaboration before briefly describing Nonaka and Takeuchi's [1] knowledge creation theory. Definitions of the various constructs of the theory are provided and three general propositions given before moving towards the operationalisation of the constructs within the collaborative R&D setting and the testing of relationships. After a presentation of the findings, we discuss the implications of the research and suggestions for future research.

2 Conceptual background

In this section we summarise the previous research on the major conceptual elements behind our study. Firstly, we mention the literature on the importance and experience with U-I collaboration. We reference the linkage between this work and the greater inter-organisational collaboration literature. Secondly, we describe briefly Nonaka and Takeuchi's [1] model of knowledge creation composed of four processes and their enabling conditions. It is important to point out that their theory is complex and interested readers should refer to the original work for a complete account of the theory's logic. Here we merely try to explain the gist of the theory and account for our operationalisation of its constructs in the context of the study conducted. We conclude with three propositions for empirical testing.

2.1 University-industry collaboration

R&D projects involving a mix of university and industry participants are ideal settings to study knowledge creation. Their explicit justification links knowledge creation to long-term competitiveness (e.g., [13]). A focus on such projects also provides internal closure [3], in terms of an end point in which successful knowledge creation is recognised and made explicit in the form of the achievement of the project's technical goals. From a practical standpoint, they are cited as important vehicles for developing a country's technological expertise and utilising and building on the knowledge and technology emanating from both university and private laboratories [14]. Government and industry champions often create organisations and programs to help foster a cooperative R&D environment.

There is evidence that collaborations involving industry participants in basic research are often not successful [15]. For example, SEMATECH, the cooperative consortium of many US semiconductor companies, failed to be fully effective in developing new process technology and moved more towards the role of creating industry standards [16]. Some cited reasons for difficulty in consortium-led collaborative R&D are the ability to appropriate any commercial benefits of the technology [17], coupled with the need for legal intellectual property protection [18] and cultural differences between university and industry settings [19]. Often, organisations close to government are formed that actively sponsor and oversee the management of such projects. In Canada, the industry-led, partially government-funded consortium of PRECARN (an acronym of Pre-competitive Advanced Research Network) has helped to form collaborative R&D projects that have produced commercially viable intelligent systems technology [20]. PRECARN recognised the problems associated with U-I collaboration early and changed its policies on intellectual property rights so that the main developer of the technology held some limited rights.

Research has examined why collaborative R&D projects are formed [21–23] and what cultural or macro-organisational factors govern their performance [14,24]. Major benefits can come from involving universities in industry R&D, which increases future industrial research and speeds up the transfer and utilisation of academic research in industry [13]. High levels of funding, a clear target, a favourable technological climate and, most notably, the selection of research projects and team members in which cooperation clearly paid off contributed to the success of a project in a case study of the

VLSI Semiconductor Research Project in Japan [25]. Despite evidence that collaboration results in positive outcomes, there is a belief that the projects are difficult to manage because of fundamental differences between university and industry settings [19]. These problems may stem from cultural aspects [26], the difficulties in development of interpersonal trust [27] and the lack of an appropriate liaison [28]. The use of collaborative policy instruments is recommended in establishing different levels of trust in some collaborative projects [14]. From the university researcher's perspective, publication assets, coordination costs, additional funding, and membership in the disciplines of engineering, natural sciences and health sciences are factors affecting the choice of becoming involved in collaborative research projects [29]. Knowledge transfer in collaborative R&D should also be seen as a two-way process of development between university and industry participants [30]. All of this suggests that successful collaborative R&D balances the interests of organisations and individuals, who differ as to their capabilities, motivations and expectations and, at the same time, maintains an open and integrative working environment. In terms of knowledge creation theory an important empirical question not answered to date is whether the theory of Nonaka and Takeuchi [1] applies in such contexts.

2.2 Knowledge creation processes

Differentiating knowledge creation from other organisational learning forms, Nonaka, Takeuchi and Umemoto defined it as "the process that organisationally amplifies the knowledge created by individuals and crystallises it as part of the knowledge system of an organisation" [3, p.833]. Because the process of technological innovation involves the creation of knowledge that is organisational, technical organisational knowledge creation can be measured as the achievement of technical objectives and used as a proxy of this particular type of organisational knowledge.

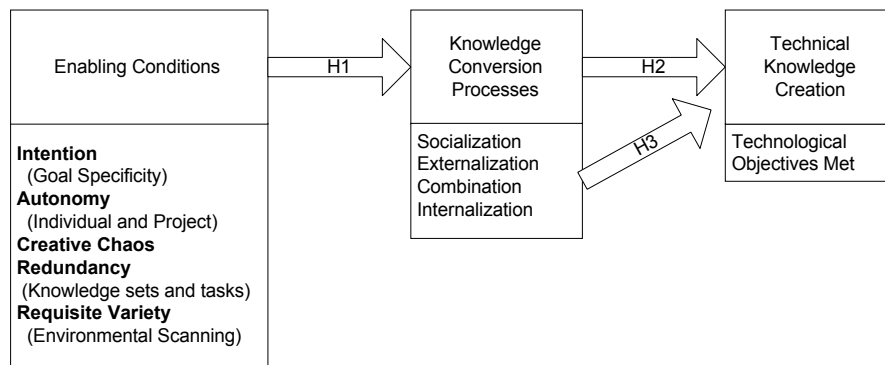
The entire model of organisational knowledge creation is, by its very nature, a multi-level process-oriented framework. Knowledge is created at the individual, group and organisational levels. In this paper, we are interested in technological knowledge created at the project level with collaborative interorganisational R&D projects. Whilst much learning and knowledge creation will take place at the individual level, we are interested in whether that knowledge was made organisational at the project level. Figure 1 is a summary of the knowledge creation adapted from Nonaka and Takeuchi's work. The four knowledge conversion processes are supported by the five enabling conditions. Both have a positive effect on project level knowledge creation, which is operationalised here as the successful achievement of technological objectives.

2.3 Knowledge conversion processes

Nonaka and Takeuchi [1] posit four knowledge conversion processes that are essential to organisational knowledge creation. According to their theory, most knowledge begins as tacit knowledge, which may reside only within an individual and only at an unconscious level. Tacit knowledge structures are verbally and often conceptually inexpressible [32,33], but play an important role in knowledge management because they constitute a vast untapped knowledge resource (e.g., [34–36]). By way of the knowledge conversion processes proposed first by Nonaka and Takeuchi [1], this tacit knowledge is made more

explicit where it may be codified and transferred among organisational members and re-internalised at the tacit level as shared or organisational knowledge.

Figure 1 Specific model of organisational knowledge creation



Nonaka and Takeuchi's [1] model of knowledge creation has been very influential in pointing out the importance of tacit knowledge in knowledge creation. The knowledge conversion processes are at the heart of the tacit/explicit dimension of Nonaka and Takeuchi's model. It is through the iterative processes in which knowledge is converted from individual tacit elements into explicit organisational elements that organisational knowledge is created. This so-called 'knowledge spiral' is created via the four knowledge conversion processes.

The first knowledge conversion process of socialisation describes the process through which people within an organisation might transfer their tacit knowledge to one another in the form of shared experiences. Socialisation processes develop individual tacit knowledge through personal experience and experimentation and perhaps through unconscious learning. Nonaka and Takeuchi name this sympathised knowledge, perhaps referencing the sympathetic nervous system, because of the characteristic that such knowledge is unconsciously transferred through shared experience.

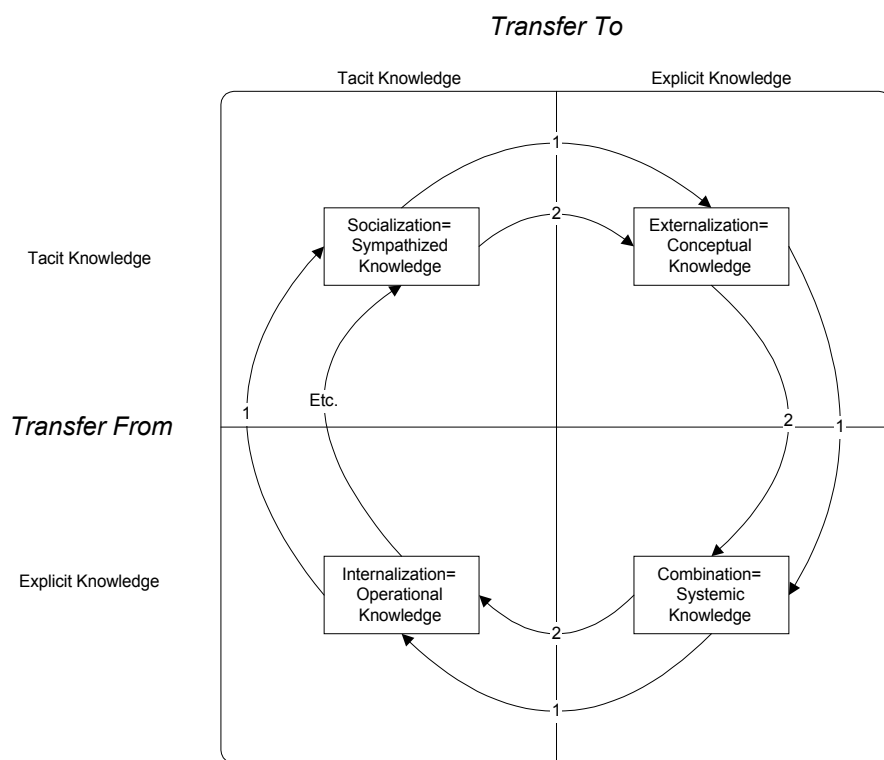
The second knowledge conversion process of externalisation describes the process of sharing tacit knowledge explicitly in the form of metaphor and analogy. This knowledge conversion process is most emphasised in Western philosophy and rational science. Mission statements and visions provided by leaders within the organisation as suggested by Hamel and Prahalad's [37] strategic intent are examples of the use of metaphor to drive organisational innovation. Nonaka and Takeuchi name this conceptual knowledge because it represents the epitome of concepts and abstraction.

The third knowledge conversion process of combination describes the process of sharing, or combining explicit knowledge. It is characterised by the linking of documentation or codified knowledge to produce something new. The use of object oriented computer programming is an example in which self-contained programs (objects) may be combined to create a new program that provides unique value-added functions. Nonaka and Takeuchi name this systemic knowledge because it may be combined as units to build larger systems that are greater than the sum of their parts.

The fourth knowledge conversion process of internalisation completes the ‘knowledge spiral’ cycle. It describes the process of knowledge creation through learning by doing or by internalising information as tacit knowledge. Apprenticeship is an example of the process at work where the student absorbs external rules, regulations or ‘tricks of the trade’ until they are internalised as belief rather than mere fact. Nonaka and Takeuchi name this operational knowledge because it involves using operational knowledge to develop a personal sense of how to do something or what something might mean to a person. Any type of skill, experience or learning by doing is relevant, such as driving a car where the student at first studies the explicit rules of the road only to internalise them later perhaps as habits, good or bad.

As illustrated in Figure 2, the processes create cycles that feed off each other in the ‘knowledge spiral’ in which each process leads to the next in a spiral unwinding through time and towards greater amounts of organisational knowledge creation.

Figure 2 Knowledge conversion processes



Source: Adapted from Nonaka and Takeuchi [1]

2.4 Five enabling conditions

Nonaka and Takeuchi also describe five enabling conditions of intention, autonomy, fluctuation and creative chaos, redundancy, and requisite variety that support the knowledge conversion processes. Intention is the ‘organisation’s aspiration to its goals...

[and] provides the most important criterion for judging the truthfulness of a given piece of knowledge” [1, p.74]. This philosophy dates back to theories of organisational decision making [38,39] and organisational sense making [40]. These and other similar concepts like strategic intent [37,41] explain what types of information and knowledge are deemed important and truthful by organisations.

Autonomy refers to the degree to which individuals, groups and business units can act on their own accord. It has been suggested that autonomy is important to creativity [42–44]. Inherently risky, autonomy may allow the flexibility needed for creative innovation. It is the concept most prominent in the use of skunk works in organisational R&D design [45–47].

Fluctuation and creative chaos represents a more complex and abstract construct in which attempts are made to create situations where crisis leads to renewal. Fluctuation is characterised by ‘order without recursiveness’ [1, p.78]. Creative chaos is fluctuation intentionally introduced by management to “(increase) tension within the organisation and (focus) the attention of organisational members on defining the problem and resolving the crisis situation” [1, p.79]. The concept refers to the intentional attempt at provoking reflective, double-loop or deutero-learning in organisational members [48].

Redundancy refers to overlap in information shared by different individuals in the firm and not to unnecessary duplication. It is “the existence of information that goes beyond the immediate operational requirements of organisational members” [1, p.80]. Redundancy allows for greater communication between organisational members who can speak the same language and help to find solutions to problems others may be experiencing. Sharing the same language or similar mindset leads individuals to understand each other better and builds on communities of practice [49,50].

Finally, requisite variety derives from Ashby’s [51] cybernetics theory in which ‘an organisation’s internal diversity must match the variety and complexity of the environment in order to deal with challenges posed by the environment’ [1, p.82]. This research focused on environmental scanning as a tractable measure of requisite variety in collaborative R&D projects. Environmental scanning will be greater for more complex projects, as per the logic of requisite variety. Environmental scanning represents one aspect of developing structures for interpreting and thus controlling the interface between the internal and external environments (e.g., [52,53]). Obtaining information on several aspects of specific environmental sectors facilitates alignment between competitive strategies and environments [53].

The theoretical underpinnings that support the link between the above enabling conditions and knowledge conversion processes are described in detail in the original work by Nonaka and Takeuchi [1] as well as in Johnson [55]. A schematic link is provided by Figure 1, which forms the basis for our major propositions.

2.5 Research propositions

Examining collaboration using the knowledge creation paradigm, we believe, provides different insights that can assist in the management of innovation in the U-I collaborative setting. In this paper we first test for the presence of the knowledge creation processes and enabling conditions predicted by Nonaka and Takeuchi’s [1] organisational knowledge creation theory. Then, we assess their impact on achieving the technical objectives of the individual U-I collaborative projects.

The model of knowledge creation examined in this research is depicted in Figure 1. The major propositions essentially address the relationship of the enabling factors and processes to technical outcomes in a linear fashion. First, the four knowledge conversion processes should be significantly related to the presence of the five general enabling conditions. This follows from the formulation of organisational knowledge creation theory [1], which stipulates that the conditions will support the processes. Therefore:

PROPOSITION 1

Knowledge conversion processes will be positively associated with each of the five general enabling conditions

Secondly, the existence of supportive knowledge processes process will impact on performance. However, no consensus exists on the definition or measurement of success in implementing technical projects. O'Connor, Parsons, Liden and Herold [56] defined four categories of outcomes:

- 1 implementation, integration and institutionalisation
- 2 human partnership dynamics
- 3 economic performance
- 4 operational effectiveness.

One 'bottom line' measure mentioned by both practitioners and academics with regards to R&D projects is the achievement of technical objectives. This measure is one of many measures used in the organisational literature that views performance as goal attainment [57]. Goal attainment is also a common measure in the R&D management literature [58]. Therefore we propose:

PROPOSITION 2

The achievement of technical objectives will be positively associated with each of the knowledge conversion processes

Proposition 2 examines the four knowledge conversion processes separately. It is expected, however, that there is a synergistic effect of the four on each other. Nonaka and Takeuchi [1] described this effect as the 'knowledge spiral'. In cross sectional research such as that described in this paper, we would expect that the relationship of an aggregate measure of the processes be greater than separate measures of each type of process. As such, we would expect an aggregate of the processes to be more strongly related to the performance measure:

PROPOSITION 3

The association between the achievement of technical objectives and the aggregated knowledge conversion processes will be stronger than the association between the achievement of technical objectives and each of the knowledge conversion processes measured separately

3 Methods

This research utilised both case studies and survey research. Six case studies were used to explore the context of knowledge creation and detect the presence of knowledge management constructs. These were conducted on-site by one of the researcher's. This provided the grounding for the items for a survey questionnaire. Then, 25 projects were surveyed by a mail/telephone survey. Given the small sample size, the collected data was summarised using descriptive and non-parametric statistics. Interesting results from this quantitative analysis were investigated by examining the transcripts of the audio taped conversations with collaborative project participants from the case studies. This leads to a better understanding of the underlying behaviours.

3.1 Research sample

The final research sample consisted of 25 collaborative R&D projects involving the creation and development of novel intelligent systems, which consist of artificial intelligence, computer and robotic technologies. The projects were all partially funded by the Canadian consortium of PRECARN, which acts as a mediator in financing and supporting these types of collaborative R&D projects. PRECARN provided access to the respondents. Each project had at least three organisational participants of three distinct types: a university or government laboratory, an industrial developer of the technology and a potential end user of the technology. Key informants for the case studies and questionnaire consisted of project leaders and other individuals (i.e., senior scientists) associated with the decision making in each collaborative project.

Prior to survey development, 21 in-depth interviews were conducted with managers from each organisation involved in six different projects. Those interviews formed the basis for the questionnaire development as well as preliminary theory development, extending Nonaka and Takeuchi's [1] model into the collaborative situation. The interviews suggested that the factors of knowledge creation in the collaborative situation were indeed often different.

After initial testing on consortium level management and similar informants from outside of the sample, the self-administered surveys were then sent to the project managers and managers from the other organisations that actively participated in the management of the 32 collaborative projects sponsored since 1990. Two of these projects ended in 1995 and were eliminated from the eligible sample due to problems in contacting informants and potential memory biases. All other projects had been completed recently, with the oldest being 18 months since completion. Two other projects were eliminated because of severe managerial problems, leading to work stoppages and cancellation by consortium management. After eliminating these four projects, the survey obtained an 89% response rate by project. For 11 of the projects (44%), multiple informants responded. The inter-rater reliability (Cronbachs alpha) for the projects with multiple informants averaged 0.76 (s.d. = 0.09; range = 0.62-0.90). The average investment for the projects was \$US 1,495,000 (Standard Deviation = \$1,360,000; Range = \$420,000-\$6,298,000). The average number of organisational participants was 4.7 (Standard Deviation = 1.7 persons; Range = 3-9 persons). Post survey analysis of non-response bias suggested that only some respondents had

knowledge of the entire project-level management processes and therefore could complete the entire survey.

3.2 Measures

The survey items from the questionnaire consisted of a seven-point Likert scale (1 = completely disagree, 7 = completely agree). Respondents were asked the extent to which conditions applied to their project. These perceptual measures were complemented by in-depth case studies of six projects. The measures for each construct are given in the Appendix. The aggregated measure of knowledge conversion processes summed the scores all of the individual knowledge conversion process measures for each project. The justification for this aggregate measure will be explained in greater detail in the results section.

3.3 Tests

In order to examine Proposition 1, correlations from the survey work were compared focusing on the connection of the enabling conditions to the processes. In examining Propositions 2 and 3, projects were clustered into high and low scores based on each of the knowledge conversion processes. This was also done with the aggregated knowledge processes score. We conducted a k-means cluster analysis using SPSS 9.0 to discriminate between high and low groupings. Non-parametric Mann-Whitney tests were then used to test for differences between the high and low groups in the achievement of technical objectives. This tests for discrimination between projects engaging in many knowledge conversion processes from those using few and the effect this has on the achievement of technical objectives. That is, does the achievement of technical objectives differ significantly for projects scoring high on any particular knowledge conversion process from projects scoring low on any particular knowledge conversion process? This is as opposed to testing only for association, as in the use of the Spearman rank correlations.

4 Results

4.1 Quantitative results

In this section we present the quantitative results. In the following section we discuss the results and present additional insights from the qualitative data that explains or expands on these findings. The Spearman rank correlations in Table 1 indicate that some, but not all, of the variables defining the enabling and process constructs were significant. The case could be made that many of the variables were not by themselves significant but interacted with others to produce significant relationships. We found that some enabling conditions contributed to knowledge conversion processes in aggregate but very little to individual processes. Similarly, knowledge processes in the aggregate contributed to the achievement of technical objectives. This is not surprising given the sample size and the preliminary stage of theory building and testing represented by this research.

Table 1 Means, standard deviations and correlations^a

Variable	Mean	s.d.	1	2	3	4	5	6	7	8	9	10	11	12
<i>Enabling Conditions</i>														
1 Goal specificity	5.44	0.88												
2 Individual autonomy	3.43	1.81	-0.38 [†]											
3 Project autonomy	4.23	1.44	0.11	0.23										
4 Creative chaos	2.09	1.28	-	0.14	-0.29									
			0.41*											
5 Redundancy in knowledge sets	3.41	1.16	0.02	-0.07	-0.05	0.22								
6 Redundancy in tasks	3.89	1.39	-0.05	-0.08	-0.07	0.26	0.57**							
7 Environmental scanning	4.48	1.43	0.42*	-0.31	-0.02	-0.14	0.12	0.00						
<i>Knowledge Creating Processes</i>														
8 Socialization	5.55	0.72	0.06	0.07	0.17	0.01	0.30	0.21	0.37 [†]					
9 Externalisation	4.50	1.29	0.09	-0.18	-0.02	0.44*	0.08	0.05	0.44*	0.38 [†]				
10 Combination	4.58	1.30	-0.15	0.39 [†]	0.26	0.54**	-0.01	0.05	-0.10	0.00	0.21			
11 Internalisation	5.78	1.03	0.09	0.53**	0.41*	-0.13	0.11	-0.24	0.04	0.39 [†]	0.05	0.10		
12 Combined KCPs	5.08	0.65	0.05	0.13	0.17	0.43*	0.12	0.11	0.33	0.68**	0.82**	0.46*	0.34 [†]	
<i>Outcomes</i>														
13 Achievement of technical objectives	5.50	1.16	0.26	-0.30	0.11	0.00	-0.04	-0.05	0.47*	0.23	0.49*	0.11	0.11	0.42*

^a N = 25 Projects; [†] p < .10; * p < .05; ** p < .01

*PROPOSITION 1**Enabling conditions and creation processes*

Intention, operationalised as goal specificity, was not correlated with the knowledge conversion processes. This was surprising given the assumption in the project management literature and management of change literature that suggests that clearly specified goals are important to managing successful projects. Creative chaos was most strongly related to the knowledge conversion processes. However, it was also found to be of low use with a mean of only 2.09 on a 7-point Likert scale. It appears to be strongly related to both externalisation ($r = 0.44$; $p < 0.03$) and combination ($r = 0.55$; $p < 0.00$). This may be due to a counteracting effect in which the use of ambiguity and chaos actually leads to greater focus on documentation and externalising ideas. This idea can be found in a quote of one manager, who stated:

“You see a lot of progress whenever you get people to write things down. [Then, you] have the ideas assessed by the whole community and work in a feedback loop where it is clear that there are changes going on in capturing the evolution of thinking about a product or a process or a prototype. [When] problems are so complex we need to be able to write them down and the more precisely we write them down the closer we are to understanding them.”

These individual processes may then manifest themselves in the organisational process of creating explicit knowledge.

The autonomy constructs were both correlated with internalisation. As such, internalisation through learning by doing can be seen to require a certain amount of autonomous action. This is a reasonable connection, as most teachers know a person cannot be forced to learn something, or more accurately innovate, against their will. The redundancy constructs were not related to any knowledge conversion process. This may also be an artifact of the sample studied. The management of most of the projects studied eschewed the use of redundant features. As one manager stated, “When you put a project together redundancy is probably your biggest problem, not your biggest asset”. Finally, environmental scanning was correlated with externalisation ($r = 0.45$; $p < 0.03$). In other words, a desire to interact with sources of knowledge, markets and events outside the confines of the project goes hand in hand with the practice of communicating with others about ideas and events within the project.

*PROPOSITIONS 2 and 3**Knowledge conversion processes and technical objectives*

The aggregate knowledge creating process was significantly correlated with the achievement of technical outcomes. However, when the knowledge conversion processes are examined separately, there are no significant correlations below a .05 level of significance. Thus, proposition 2 was not supported but proposition 3, when all four of the knowledge conversion processes are accounted for, was supported. This finding seems to demonstrate the importance of the knowledge spiral in affecting organisational knowledge creation. That is, it agrees with Nonaka and Takeuchi's [1] theory in which all four knowledge conversion processes act in terms of an ongoing spiral of activity. Due to the cross-sectional nature of the quantitative data, such a finding is not surprising. During the ongoing process of knowledge creation any one particular knowledge conversion process will predominate and particularly in technical knowledge creation externalisation

will prevail at the end of a successful project. The qualitative findings of the research, which are more indicative of a process model, also support this framework and are discussed in the discussion section.

Table 2 presents the results of the Mann-Whitney tests. One near significant difference at the 0.10 level of significance (i.e., .06) between low and high scoring projects was that the higher achieving projects tended to have a higher reported involvement with externalisation processes.

Table 2 Tests on achievement of technical objectives^a

	<i>Socialisation Processes</i>	<i>Externalisation Processes</i>	<i>Combination Processes</i>	<i>Internalisation Processes</i>	<i>Knowledge Conversion Processes</i>
High scoring cluster (n)	17	12	14	24	14
Low scoring cluster (n)	8	13	11	1	11
Mann Whitney U	46.5	43.5	65	4.5	40.5
Exact significance	0.215	0.060 [†]	0.536	0.400	0.044*

^a N = 25 Projects

[†] p < .10

* p < .05

5 Discussion

Overall, the results of the research suggest that some enabling conditions of organisational knowledge creation theory do support the knowledge conversion processes, which in turn have a positive effect on the achievement of technical objectives. Of the four knowledge conversion processes externalisation processes were particularly related to successful knowledge creation as measured by achievement of technical objectives. Nonaka and Takeuchi [1, p.198] stated that “in terms of knowledge-conversion modes, the Western strength lies in externalisation and combination” and thus predicted this finding. The general process of converting an idea into a patent may be considered an extended process of externalisation. Given the preoccupation of firms with patenting of technical work it is no surprise that this knowledge conversion mode is closely linked to the achievement of technical objectives.

The aggregate effect of the knowledge conversion processes was positive on achieving technical objectives. This suggests that all of the modes do contribute to the knowledge conversion process when their effects are examined synergistically. The overall project effects of individual processes are difficult to measure, as it appears that each acts temporally across any particular project. In fact, the theory suggests that processes of socialisation begin the knowledge creation processes of the project, which are then followed by externalisation processes. Examination of the case study data indicates that socialisation processes are, in fact, often utilised at the beginning of projects. One project manager made it a point to have a ‘get to know each other’ meeting

at the beginning of the project and this is a common R&D project management technique in which early phases are associated with greater face-to-face communications (e.g., [59]). In another project developing scheduling technology for steel production, members from the project who were experts in computer systems design, scheduling and constraint-based algorithms visited facilities at Chaparral Steel in Texas, one of the industry partners, to learn first hand the process of making steel from production workers and management. However, whether one or another knowledge conversion process is dominant at different phases of a project or for certain kinds of technical work or environmental conditions must await further study. These new studies will need to focus on individual knowledge processes and follow their execution over time to develop a deeper understanding of the temporal effect.

The cross-sectional correlations in this study between isolated pairs of variables also provide interesting fodder for future work. For example, internalisation was most associated with individual autonomy. This makes sense as individual autonomy allows exploration that leads to the internalisation of experience as knowledge. If that knowledge is later converted into explicit knowledge through the knowledge conversion processes it may lead to valuable project knowledge being created. However, there was anecdotal evidence from the qualitative data that individual autonomy might be detrimental to the overall project objectives. For example, one manager stated:

“It was more a question of communication. You could say that if you don’t have communication then you have autonomy. If you have several groups that are supposed to be part of a larger group moving off in the wrong direction...[that’s bad autonomy].”

Another project leader stated:

“There is a lot of autonomy on this project. Is that the reason why we’re better or worse at creating knowledge? I don’t know. People like autonomy and their own space to create, but if it’s not done within a structure, you can get results that are not what you want.”

Thus, if internalisation and individual autonomy are associated with individual knowledge exploration that is not project-related, such knowledge creation may not feed back into the project as a whole and will not become ‘organisational’. The time and energy spent on individual avenues of exploration may then have a negative effect on project-level knowledge creation.

This research also suggests that the specific enabling conditions and knowledge conversion procedures utilised depend on the organisational context (i.e., single versus multiple organisations) in which they are to be utilised. Nonaka and Takeuchi’s [1] data was primarily based on single organisation contexts where ours is based on multiple organisation contexts. For example, the potential problems associated with individual autonomy discussed above may be less apparent within a single organisational context. In that case, individual exploration may have a greater chance of leading to further learning that is incorporated into the organisational knowledge system of the firm as long as the individual is still employed at the firm. This supports the notion of practices in companies such as 3M, which allows its employees to devote 15% of their time to individual exploration on company time [60,61]. The learning of such individual explorations often flows back into the firm by socialisation processes. The transitional nature and specific objectives of most inter-organisational projects, however, makes the probability of individual exploration leading to project level learning less likely.

The overall findings of the research suggest that organisational knowledge creation factors and processes are relevant in the U-I collaborative R&D project context. However, not all of the factors were significant in this context and the knowledge conversion processes were only significant when their aggregated effects were analysed. Statistically, this suggests that the aggregate (conceptualised as the original knowledge spiral proposed by Nonaka and Takeuchi [1]) is a better predictor of knowledge creation than the separate measures of knowledge conversion processes. These findings provide direction to future research in the area of organisational knowledge creation. They suggest that it is the entire chain of processes that is important, as the theory would suggest, and that individual processes are less relevant unless they are tied into the larger knowledge spiral of the original theory.

5.1 Implications and future work for practice and theory

For practising collaborative project managers, the major implication of this study is that a balance is required in the management of the activities of project participants. There is probably no one lever that unlocks successful outcomes but rather a series of levers pulled at appropriate times that maintain a cycle between external scanning and internalisation. Furthermore, internalisation processes may not always be beneficial. Internalisation processes are counter-productive when individual learning is internalised but not eventually shared. In collaborative projects this suggests cross-fertilisation mechanisms that to some extent dissuade individual autonomy but at the same time require internalisation. If the building of organisational memory is desirable then it requires that there are processes in organisations to take useful knowledge and retain it over time. These processes are social in nature and act through structures such as interpersonal networks (e.g., [62]). One major role of management and innovation facilitators such as PRECARN is to hold knowledgeable people accountable for engaging others in dialogue versus insular studies.

Collaboration between two organisations with similar goals and environments (i.e., for-profit and in the same industry) can be problematic. Even more difficult can be two organisations from different cultures. Most relevant to the U-I context in this paper are the differences in culture between academics and industrialists. Applying some of the results from the larger research program described here to future research creates the possibility of expanding the usefulness of knowledge creation theory to collaborative contexts. For example, the detrimental effects of a lack of integration between business and research goals were examined in two in-depth case studies of projects, one successful and the other not, managed by the same project leader [63]. When the goals and cultural background were compatible the project was successful and when it was not, the project was unsuccessful. Work on practitioner-academic research teams suggests that each group comes to the project with different goals such as peer reviewed publications for academics and commercial results for practitioners [64]. Whilst these individual goals may be different they are reconcilable in the knowledge creation processes, if the projects are managed properly. Also, funding in collaborative research between academics and practitioners is considered to be underutilised even though it is expected to increase [65]. More work must be done in relating context to knowledge creation and ultimately performance.

Concerning theoretical implications, the growing literature on interorganisational relationships leads us to believe that the dynamics of interorganisational projects differ substantially from the dynamics of single organisation projects. Therefore, expanding the current single organisation project perspective to an interorganisational perspective is a valuable contribution to knowledge management theory. This research is unique in that it explores a collaborative interorganisational context.

The different knowledge conversion processes appear interrelated and necessary, albeit on different levels, in achieving the overall success of project-level knowledge creation. This would agree with Nonaka and Takeuchi's [1] original concept of the knowledge spiral. However, larger research samples and delicate temporal research designs need to be utilised in future research. One of the advantages of a larger sample of collaborative projects would be the possibility of performing a structural equation model on a rationalised model composed of enablers from the four knowledge processes and a more complex performance construct embodying multiple measures of success.

Future research looking at projects from two or more homogenous within-group projects would shed light on the presence and effects of organisational knowledge creation factors and processes in different ethnic cultural settings. Lam [66] has already demonstrated the problems that materialise when looking at between-group projects that involve organisational members from two different ethnic cultures. Future research may also focus on the structure of knowledge processes and the presence of enablers in reconciling ethnic cultural differences.

6 Conclusion

Collaborative R&D management, where knowledge creation must occur at the inter-organisational project level, is difficult and not well understood but important to technical progression [11,12]. This research examined the effects of enablers and processes of knowledge creation in a collaborative setting. It demonstrates the importance of some of the enabling conditions to the underlying process of knowledge conversion modes defined in the Nonaka and Takeuchi's knowledge creation theory and the overall effect of these knowledge conversion process modes on achieving technical objectives.

The most important findings are that the knowledge conversion processes are most relevant when examined together as opposed to separately and that the enabling factors and processes appear to be contingent on the organisational context studied (i.e., single versus multiple organisations). Whilst the underlying combined effects of the knowledge conversion processes, when measured as an aggregated score for each project were strong in the interorganisational setting, their influence when measured singly was negligible. Given the cross-sectional nature of the data, this agrees very much with the theory proposed by Nonaka and Takeuchi [1], that it is the knowledge spiral process itself that is important rather than any individual knowledge conversion process on its own.

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Appendix

Constructs and measures used in the survey analysis

Dependent variable: achievement of technological objectives

Scale alpha was 72%. Specific items were:

- overall, the project was successful in meeting most of its original goals
- all of the original technological objectives were met.

Independent variable: goal specificity

Scale alpha was 80%. Specific items were:

- there were specific, well-defined goals established at the beginning of the project
- a major managerial undertaking was the establishment of project goals
- the initial goals of the project helped to define the majority of the tasks that needed to be accomplished in the project
- one definite intention of the project was the creation of new knowledge
- the project's initial goals helped drive the innovation in the project
- without making the initial intent of the project explicit, innovation would not be possible.

Independent variable: individual autonomy

Single item scale. Specific item was:

- within the project, individuals often went beyond the boundaries of their assigned tasks to work on whatever problem task they desired.

Independent variable: project autonomy

Single item scale. Specific item was:

- the project's management was autonomous, in that the management of the organisations making up the project team did not govern its policies and procedures.

Independent variable: creative chaos

Scale alpha was 90%. Specific items were:

- management intentionally created at least one artificial crisis within the project to stimulate innovation.
- to stimulate innovation, management intentionally introduced ambiguity regarding tasks and their allocation within the project.

Independent variable: redundancy in knowledge sets

Scale alpha was 64%. Specific items were:

- if the project lost one of its participant members the original goals could still be achieved with knowledge held by the other participants
- the participants had too much information to deal with from the project
- there was a fair amount of intentionally built-in overlap in information between project participants.

Independent variable: redundancy in tasks

Single item scale. Specific item was:

- two or more groups within the project worked on the same tasks simultaneously to stimulate innovation through competition.

Independent variable: environmental scanning

Scale alpha was 87%. Specific items were:

- project management actively measured the market for technology
- project management actively measured the industry aspects of the external environment (e.g., potential competitors)
- project management actively measured aspects of government regulation
- project management actively measured political aspects of the external environment (e.g., ethical implications of technology)
- project management actively measured social aspects of the external environment (e.g., environmental implications of technology).

Independent variables: knowledge conversion processes

Scale alpha for the aggregated knowledge conversion processes construct was 75%.

1 Independent variable: socialisation

Scale alpha was 68%. Specific items were:

- we brainstormed ideas
- face-to-face meetings were used as important sources of ideas
- face-to-face meetings were good for developing a shared understanding between members of the project
- face-to-face meetings helped develop shared technical skills.

2 *Independent variable: externalisation*

Scale alpha was 83%. Specific items were:

- written documents were used to express new ideas
- vivid imagery was used to express new and unfamiliar ideas with other members within the project
- written documents were used to increase learning in the project.

3 *Independent variable: combination*

Scale alpha was 47%. Specific items were:

- the project created a new concept by combining two or more already familiar concepts
- two or more participants produced documents in isolation that were later combined into a larger single document

4 *Independent variable: internalisation*

Single item scale. Specific item was:

- learning by doing was used as an essential element of the project.